

A numerical model for evaluation power outages impact on water infrastructure services sustainability

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Abstract

Critical infrastructure's (CI) (electricity, heat, water, information and communication technology networks) security, stability and reliability are closely related to the interaction phenomenon. Due to the increasing amount of data transferred, increases dependence on telecommunications and internet services, the data integrity and security is becoming a very important aspect for the utility services providers and energy suppliers. In such circumstances, the need is increasing for methods and tools that enable infrastructure managers to evaluate and predict their critical infrastructure operations as the failures, emergency or service degradation occur in other related infrastructures. Using a simulation model, is experimentally tested a method that allows to explore the water supply network nodes the average down time dependence on the battery life time and the battery replacement time cross-correlations, within the parameters set, when outages in power infrastructure arise and taking into account also the impact of telecommunication nodes. The model studies the real case of Latvian city Ventspils. The proposed approach for the analysis of critical infrastructures interdependencies will be useful for practical adoption of methods, models and metrics for CI operators and stakeholders.

Introduction

The safety, security and reliability of Critical infrastructures (CI) are strongly governed by mutual interaction phenomena. Direct dependency mechanisms are relatively easy to identify, model and analyze, however in the case of multiple, large-scale critical infrastructures, direct dependencies between elements form loops, and give rise to mutual dependencies, i.e., interdependencies [1].

In the last years, CI interdependency issues have been discussed widely among researchers. Zimmerman et al. proposed explicit metrics to quantify interdependencies [10], [11]. In [10] the author offers metrics for cascading effect in the context of electric power such as Relative duration of the cascade. A MICIE project among other objectives considered methodologies, algorithms and tools to perform quantitative evaluations of risks and threats deriving from interdependencies existing among CIs [3]. Casalicchio et al. in his paper [6] presented a taxonomy that classifies interdependency metrics based on their information content, decision support and risk analysis capabilities, and computational costs. A structured view of dependability according to the threats, i.e., faults, errors and failures, the attributes and the means for dependability is presented in [7]. Bagheri [8] proposed an extension to the Unified Modeling Language (UML) in order to define a model for research of CI dependences and a fair basis for multi-agent modeling and simulation of critical infrastructures. UML multi-agent model that captures the static structure and dynamic behavior of a water distribution networks was presented in [9]. In the MIA project [12] CI models were classified according data availability

(granularity): Topological model, Analytical model and Simulation model. Furthermore, the MIA project developed new CI interdependencies metrics and applied them in topological, analytical and simulation models. A simulating model of three CI interdependency, which applies by MIA project developed metrics, is presented in [16].

Due to the increasing amount of data transferred, increase a dependence on telecommunications and internet services, the data integrity and security is becoming a very important aspect for the utility services providers and energy suppliers. In such circumstances, the need is increasing for methods and tools that enable infrastructure managers to evaluate and predict their critical infrastructure operations as the failures, emergency or service degradation occur in other related infrastructures.

The contribution of this research is:

- To describe how the simulation approach allows to model in deeper details critical infrastructures coupling and their behavior.
- To develop a MATLAB® Simulink® State Flow simulation model of a modern water supply system (composed of water supply nodes, telecommunication nodes and power system nodes).
- To simulate the impact of effect of power nodes outages on telecommunication and water supply nodes and to assess the impact of failure propagation, using metrics developed in previous researches.
- To describe dependences between infrastructures by polynomial and to minimize impact on water node down time.

The novelty of this research lies in the fact that using a simulation model is experimentally tested a method that allows to explore the water supply network nodes the average down time dependence on a battery life time and a battery replacement time cross- correlations, within the parameters set, when outages in power infrastructure arise and taking into account also the impact of telecommunication nodes. The model studies the real case of Latvian city Ventspils.

The approach described in this work will help service providers allocated in neighborhood to plan risk mitigation actions. For example, power supply operator being aware about impact on telecommunication or water supply CI could prioritize restoration actions in case of emergencies, or water supplier could make a decision about spare power supply for water distribution network nodes.

The paper is structured as follows. The metrics for dependencies evaluation are presented in a Section 1. An example of dependence of water supply nodes from the outages happened in power grid and telecommunication nodes is described in Section 2, which details main steps proposed by authors for modeling and simulating the state of critical infrastructures. The main contribution of the research, general results and possible directions for future work are discussed in Section 3.

1. Metrics for dependencies evaluation

In order to assess dependencies between infrastructures nodes the authors selected the metrics defined in the MIA project [12]: *Ratio of physical dependencies* and *Temporal scale of inter-dependencies*.

The purpose of ratio of dependency mentioned above is to assess:

- The direction of the dependencies (and therefore the presence of interdependency in case of bidirectional dependencies);
- The strength of the coupling versus the direction.

The interdependency between two infrastructures may be indirectly assessed by the combinations of the metrics.

The ratio of dependency is computed as follow:

Ratio of physical dependency water infrastructure:

$$RPD_{WN} = \frac{N(WN | PN)}{N(PN)}, \quad (1)$$

where $N(PN)$ is the rate of malfunction in the power nodes and $N(WN|PN)$ is the rate of malfunction in the water nodes due to loss of power system.

Ratio of physical dependency telecommunication infrastructure:

$$RPD_{TN} = \frac{N(TN|PN)}{N(PN)}, \quad (2)$$

where $N(TN|PN)$ is the rate of malfunction in the telecommunication nodes due to loss of power system.

$N(PN)$ should be evaluated as the number of times in a time interval T the performance of the power infrastructure or its component degrades due to an external events.

When we consider the ratios (1) and (2), we can assume a value greater, less or equal to 1. Let us suppose that $N(WN|PN)/N(PN) \geq 1$. It means that water nodes have a strong dependence on power supply nodes. Vice-versa if $N(WN|PN)/N(PN) < 1$ means that water nodes has a weak dependence on power supply and more the ratio goes to zero, weaker is the dependence.

Temporal scale of interdependencies:

Metrics were developed for taking into account the temporal characterization of the inter-dependencies examine the ratio of the duration of an outage of one infrastructure to the duration of the cascading outage of another impacted infrastructure [11][12].

The concept of *Temporal scale of interdependencies* can be used in a scenario in which the electricity, telecommunications and water supply network interaction, in order to measure *Temporal scale of physical interdependencies (TFI)* and *Temporal scale of cyber interdependencies (CFI)*.

Water supply nodes *Temporal scale of physical interdependencies:*

$$TFI_{NW} = \frac{T(NW)}{T(NP)} \quad (3)$$

Telecommunication network nodes *Temporal scale of physical interdependencies:*

$$TFI_{NT} = \frac{T(NT)}{T(NP)} \quad (4)$$

Power supply network nodes *Temporal scale of cyber interdependencies:*

$$TCI_{NP} = \frac{T(NP)}{T(NT)} \quad (5)$$

If the ratio in (3), (4) and (5) is less than 1, it means that the dependent infrastructure has self-healing properties, that recovery or protection mechanisms from the faults are very fast, that is it is capable to rapidly recover from the failure by themselves and then that it is capable to mitigate the dependency effect. On the contrary if the ratio (3), (4) and (5) is greater or equal than “1” the dependent infrastructure is strongly coupled with the source of failures.

2. Interdependence model development and simulation

In last few years several Automated Meters Reading and smart metering systems have been installed at water distribution networks (WDN) and district heating (DH) networks in Latvian municipalities in order to reduce water leaks and heat losses, to improve billing and customer services and to enhance efficiency of DH services.

In 2013 under a project “Smart Metering” [15], a wireless sensor network was developed at a limited trial WDN segment in Ventspils. The territory under consideration comprises 1 power substation, 1 GSM base station and 24 WDN nodes equipped with sensors- transmitters that use GSM-GPRS networks for metering data transmitting. Because the telecommunication infrastructure is widely utilised for water flow, water pressure and other data meter collection, we decided to measure and evaluate outages and fault propagation impact occurred in power and telecommunication nodes on the water distribution’s network nodes.

The proposed simulation scenario describes water distribution network nodes interaction with electricity and telecommunications network nodes.

An assumption is made:

- A model allows to explore the water supply network nodes the average down time dependence on the battery life time and the battery replacement time cross- correlations, when outages in a power infrastructure arise and taking into account also the impact of telecommunication nodes.
- It is possible to examine the interaction with a polynomial, and minimize the water supply network nodes the average down time within bound parameters.

$$R_{i,j,k} = f(t, m_j^1, m_j^2, \dots, m_j^p, m_k^1, m_k^2, \dots, m_k^p, m_i^1, m_i^2, \dots, m_i^r) \rightarrow \min, \quad \text{where}$$

$$m_j^l \in M_j; m_k^n \in M_k; m_i^r \in M_i$$

$R_{i,j}$ - function $f(\cdot)$ time t . A specific set of metrics: M_j (power supply network), M_k (telecommunication network) and M_i , (water supply network), used to measure infrastructures j , k and i performance level.

2.1. Model parameters justification

Water distribution network nodes parameters selection:

Backup battery life time lifetime (BLT): based on a preliminary analysis of products available on the market. The selected water supply network nodes running time of the backup power supply (BLT) is assumed 0.1 - 6 hours.

Backup battery replacement time (BRT) is assumed 4 – 48 hours [15].

Power supply network nodes parameters selection:

Parameters of power supply node were selected by analysing the SAFI, SAIDI and CAID data of Talsi and Ventspils distribution network regions and total data about Latvia of 2011 [13]. Model parameters are calculated using the above data (see Table I).

Table I: Model parameters: mean time to repair and mean time to failure

Parameters (hours)	Talsi	Ventspils	Latvija
Average mean time to repair (MTTR average)	2.00	1.98	2.81
Maximum mean time to repair (MTTR max.)	6.05	7.02	Not available
Average mean time to failure (MTTF average)	43817	66235	1564
Minimum mean time to failure (MTTF min.)	13519	20278	Not available

As a result, the model uses the following parameters:

- *Mean time to failure*: 1564 : 5 = 313 hours. (200 – 500 h.)
- *The simulation time* is limited 4000 hours that enables 9-12 cycles.
- *Mean time to repair*: 2.81 hours (0.28 – 5.34 h.)

Telecommunications (GSM) network nodes parameters selection and justification

Mean time to failure (MTTF): the model simulation results are used.

Backup battery life time lifetime (BLT) - according to the regulations of telco operator [14] TELE 2 base station backup power supply length is designed for 2 hours. The assumption was made that the minimum and maximum duration to be used in the model: 0.1 - 5 hours.

Mean time to repair – TELE 2 for backup power sources uses portable generators. A pessimistic assumption is done that the replacement of a battery or connection of a portable generator can last from 3 to 32 hours.

2.2. Model description

This study uses MatLab® Simulink® StateFlow modelling and simulation tool as a popular engineering and scientific modelling tool. Fig. 1 depicts the system model, which includes a power supply node, a telecommunications network node and three WDN nodes.

In addition, Fig.2 shows a part of simulation model representing the water node operation.

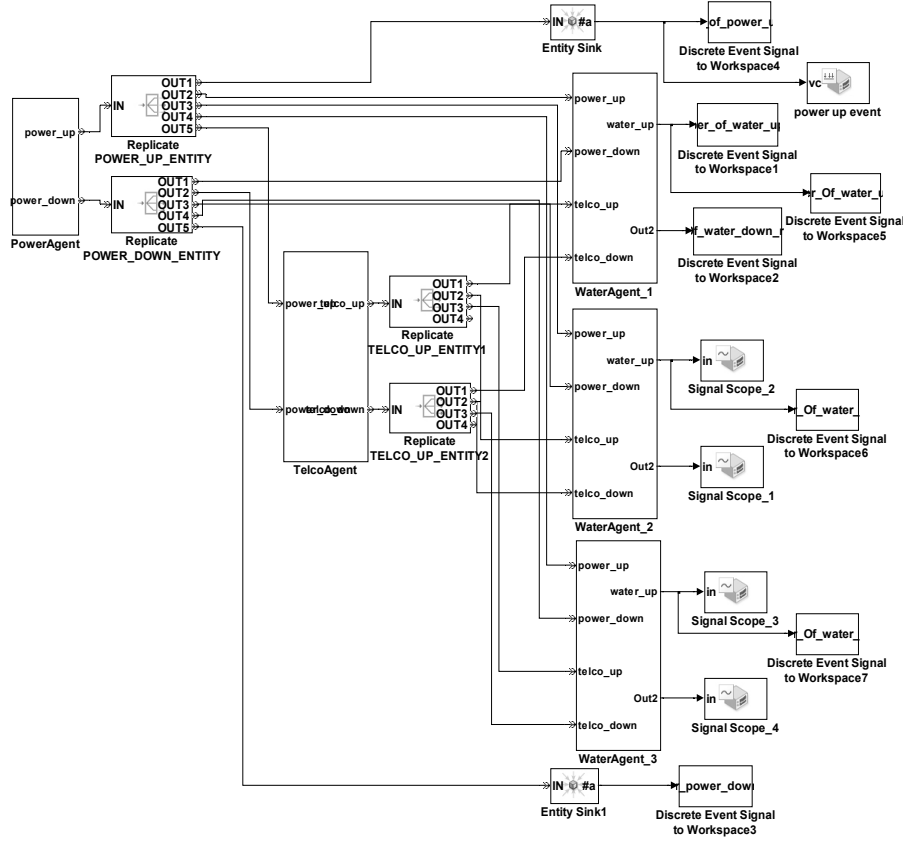


Fig.1: The system model, which includes power supply, telecommunications and water supply network nodes

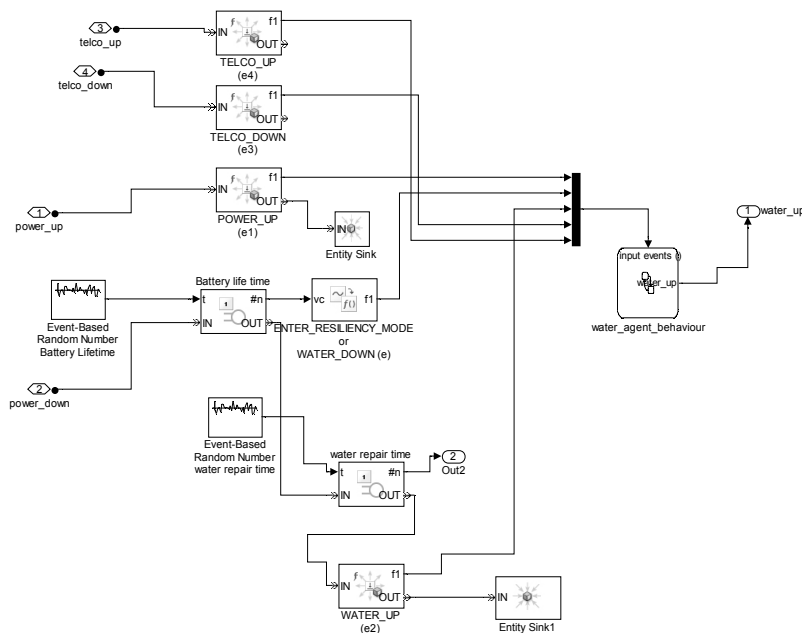


Fig.2: A part of the model representing the water node operation

Water inputs to node (see Fig. 2) are provided with exits from the power supply model: average downtime ("power_down") and mean time between failures ("power_up"). Inputs from the telecommunications network node model comes in the form of downtime ("telco_down") and as the restoring signals ("telco_up"). The first random numbers generator generates BLT, and the second random number generator generates BRT parameters.

"State flow chart" (SFC) at Fig. 3a switches between water node states "water up", "water node in resilience mode" (receives power from the backup source) and "water down", when the node is not working in normal mode due to power supply or telecommunications infrastructure failures. Fig. 3b shows the interaction between inputs, functions and ports of the SFC water node.

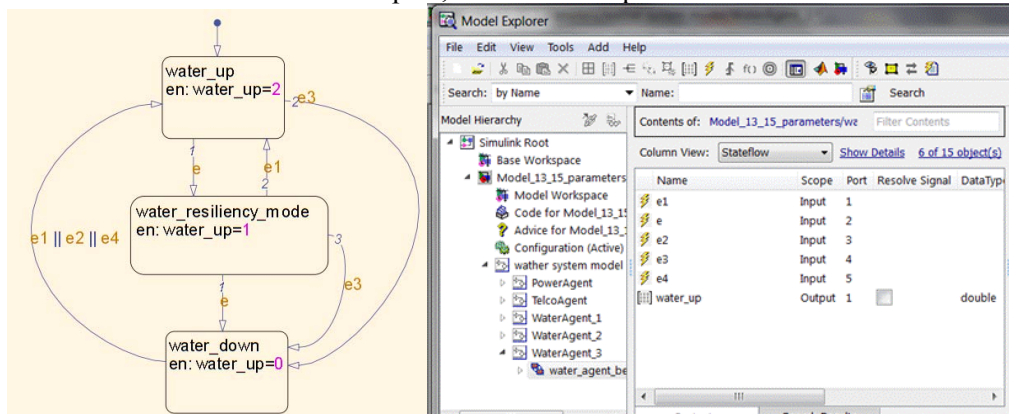


Fig. 3: a) Water node „State Flow Chart”; b) Model Explorer, which shows the interaction between inputs, functions and ports of water node SFC

Figure 4 depicts the number of power nodes downtime, while figures 5 and 6 show the water supply network node state. A state "2" means the normal operating mode, "1" – a node provides its functions using a backup power source, and "0" – a node is not functioning due to the lack of power, or because a telecommunications network node is not available.

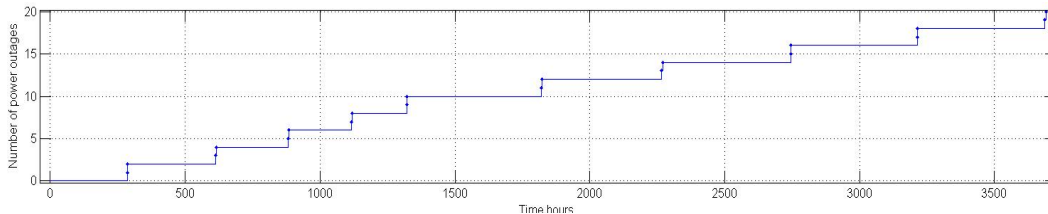


Fig. 4: Power supply node downtime during the simulation

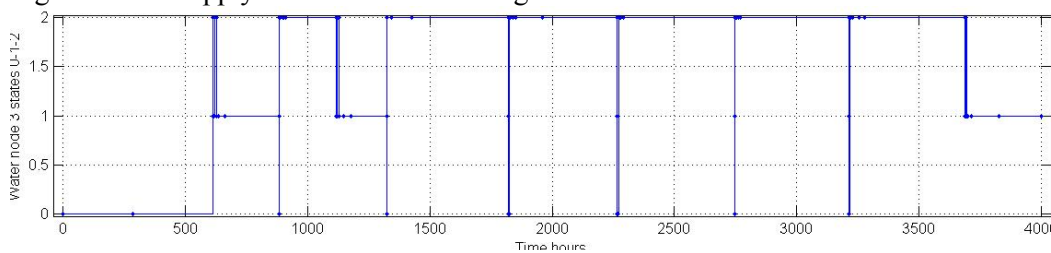


Fig. 5. Water supply network node (W1) downtime during simulation: 0-1-2

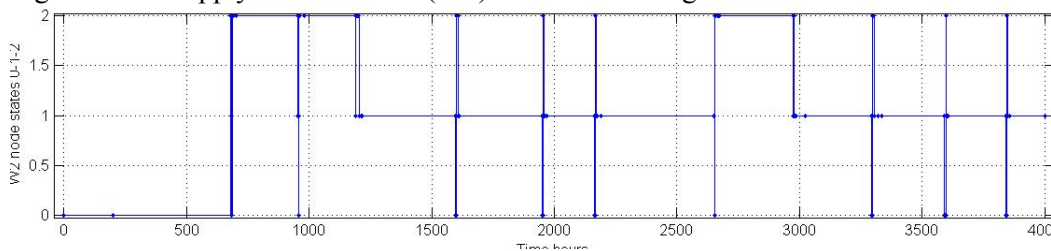


Fig. 6. Water supply network node (W2) downtime during simulation: 0-1-2

The model simulation results are summarized in the Table II. Simulation results show the close relationship between the power supply nodes failures and downtimes in water distribution networks, because: *Ratio of physical dependency*

$$RPD_{NW} = \frac{N(NW | NP)}{N(NP)} = 116 / 150 = 0.77$$

WDN average downtime 2.36 hours and power supply nodes average downtime is 2.69 hours, so the time scale of physical dependency ratio: *Temporal scale of physical interdependencies*:

$$TPI_{NW} = \frac{T(NW)}{T(NP)} = 0.88$$

If one uses average water nodes downtime, given the number of power network nodes downtime, 1.80 hours, then:

$$TPI_{NW} = \frac{T(NW)}{T(NP)} = 0.67$$

Table II: The model simulation results summary

Simulation results	Simulation runs							
Simulation numbers	1	2	3	4	5	6	7	8
Water distribution network backup battery lifetime (h)	0.1	3	6	1.5	4.5	5.5	2.75	1
Water distribution network backup battery replacement time (h)	48	24	4	36	12	46	43	38
Water distribution network average downtime (h)	2.27	2.34	1.49	1.81	2.12	2.65	2.14	1.75
Power supply network average downtime (h)	2.34	2.34	2.34	2.55	2.55	2.55	2.49	2.49
Temporal scale of physical interdependencies of water nodes	0.97	1.00	0.64	0.71	0.83	1.04	0.86	0.70
Water distribution network average downtime, given the number of power network downtimes (h)	2.27	1.56	0.99	1.92	1.65	1.77	1.71	1.81
Temporal scale of physical interdependencies of water nodes, given the number of power network downtimes (h)	0.97	0.67	0.42	0.75	0.65	0.69	0.69	0.73

Simulation results	Simulation runs							
Simulation numbers	9	10	11	12	13	14	15	Average
Water distribution network backup battery lifetime (h)	4	2.5	1.75	5	0.5	3.5	2	
Water distribution network backup battery replacement time (h)	32	28	22	19	15	10	6	
Water distribution network average downtime (h)	2.88	3.16	1.74	3.62	2.42	2.60	2.46	2.36
Power supply network average downtime (h)	2.49	3.16	3.30	3.30	2.80	2.80	2.80	2.69
Temporal scale of physical interdependencies of water nodes	1.16	1.00	0.53	1.10	0.86	0.93	0.88	0.88

Water distribution network average downtime, given the number of power network downtimes (h)	1.73	2.01	1.74	1.98	2.42	1.82	1.57	1.80
Temporal scale of physical interdependencies of water nodes, given the number of power network downtimes (h)	0.69	0.64	0.53	0.60	0.86	0.65	0.56	0.67

The experimental results demonstrate not only the lack of an escalation effect, but also the WDN significant reduction of downtime, because the temporal scale of physical dependency ratio, taking into account the power supply node downtime number, is 0.67

2.3. The polynomial approximation of the experimental results

Simulation data have been processed by EDAOpt experimental results processing tool developed by researchers of RTU [17].

The analysis of the WDN nodes average downtime ratio dependence on backup battery lifetime (BLT) and backup battery replacement time (BRT) cross-correlation parameters specified parameters is made. Fig. 7 depicts two-and three-dimensional images that demonstrate the correlation between the water WDN nodes BLT, BRT and WDN nodes downtime.

EDAOpt computed the polynomial coefficients of the quadratic approximation with one attempt exclusion.

$$R = 2.3532 - 0.3894 \cdot X_1 + 0.00812 \cdot X_2 + 0.0289 \cdot X_1 \cdot X_1 + 0.0035 \cdot X_1 \cdot X_2$$

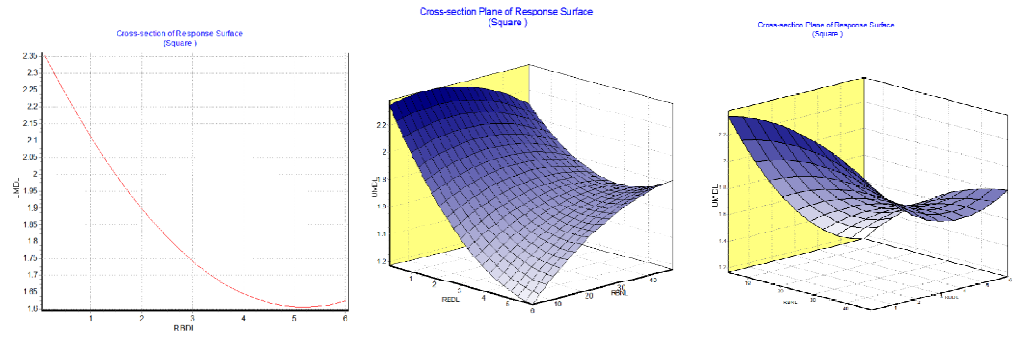
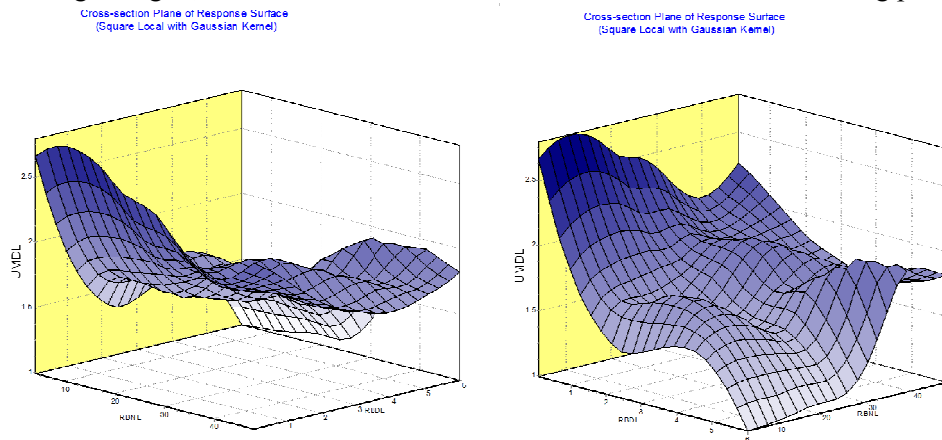


Fig. 7: Quadratic approximation with one attempt exclusion

The following local approximation is carried out with the aim of identifying possible local minimis. Looking at Fig. 8, one can see at least two local minimums within modelling parameters set.



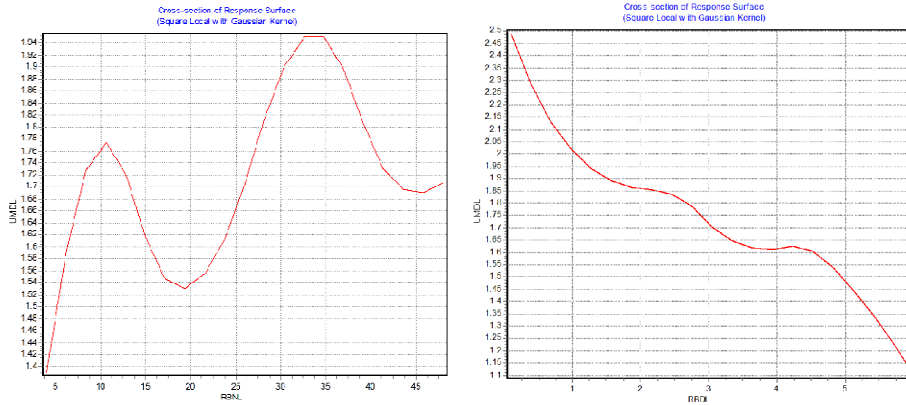


Fig. 8: Local quadratic approximation with one attempt exclusion

3. Conclusions and future work

Using *MatLaB*[®] *Simulink*[®] *Stateflow* simulation model is designed and experimentally tested a method, which, unlike the existed methods, allows to explore the water supply network nodes the average down time dependence on the battery life time and the battery replacement time cross-correlations, within the parameters set, when outages in power infrastructure arise. The impact of outages in telecommunication nodes also is taking into account. It has been proven that:

- Water nodes down time shows by a quadratic polynomial described the dependence of the backup battery life time lifetime (BLT) and backup battery replacement time (BRT) cross-correlation in parameters set;
- Although approximations graphs shows that in the explored parameter range polynomial function approaches to a minimum when BLT tends to peak, however, local minimums of the function are revealed, which are associated with a telecommunications network nodes resilience to external power supply interruptions;
- Explored that backup power sources for water supply network nodes is an effective solution, taking into account the reasonable costs, however, it is not sufficient to ensure consistent and reliable data collection, because the physical dependency ratio is 0.77 , but the time scale of physical dependency ratio is 0.67 ;
- The solutions that ensure consistent and reliable data collection from the water supply network objects can be: backup power source lifetime increasing or / and additional (redundant) network node installation that in this way increase the sensor network robustness.

Optimization of parameters of the infrastructure nodes such as batteries lifetime and meantime to restoration on the territory under consideration should be a topic of a future research. Probably the common measure such as the cost, related to considered network parameters, should be used in order to find a balance between the efforts needed for restoration of impacted infrastructure and the cost related to power back up of the affected nodes.

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